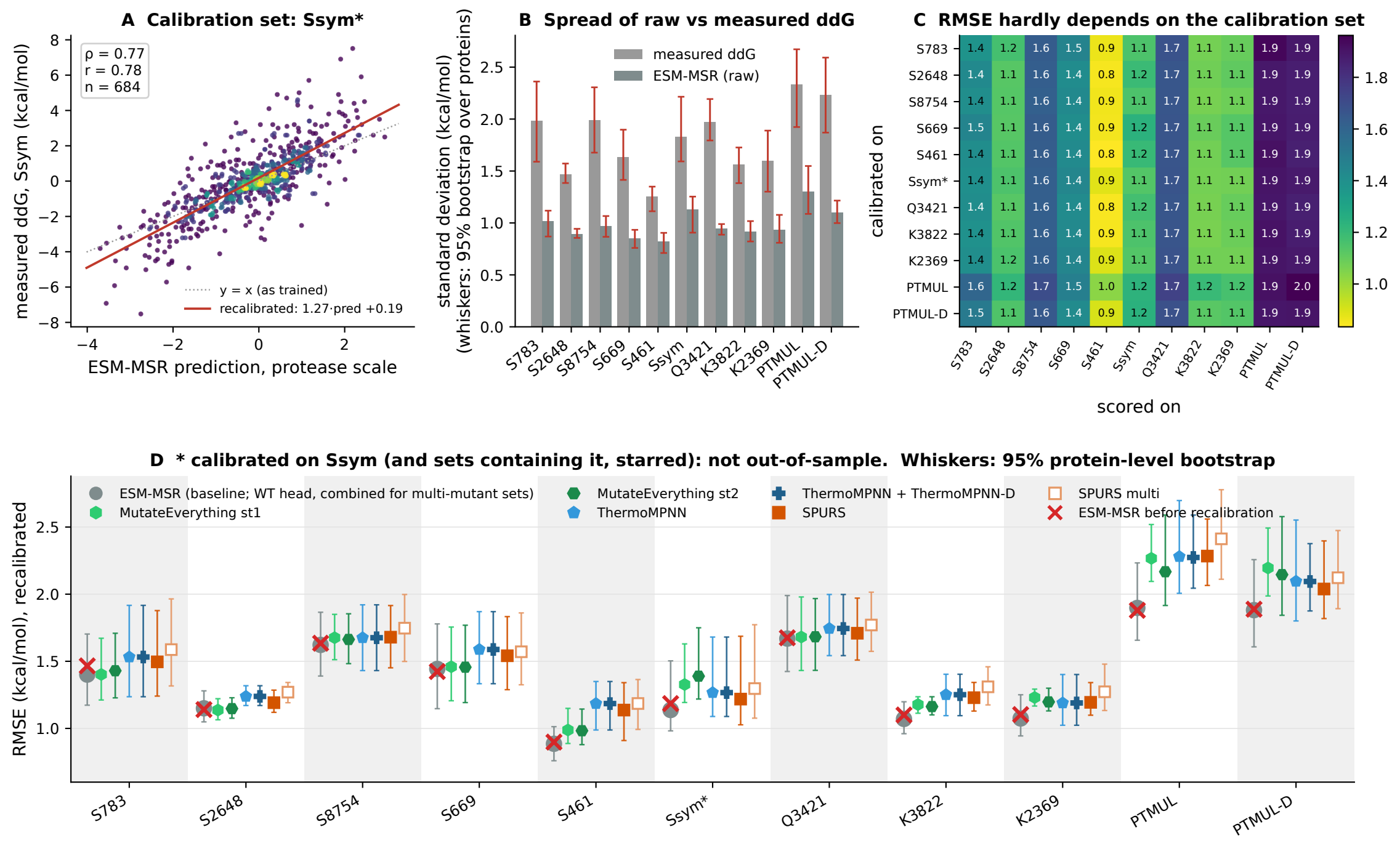

A Calibration set: Ssym*
ρ = 0.77
r = 0.78
n = 684
y = x (as trained)
recalibrated: 1.27·pred +0.19
measured ddG, Ssym (kcal/mol)
ESM-MSR prediction, protease scale
B Spread of raw vs measured ddG
measured ddG
ESM-MSR (raw)
standard deviation (kcal/mol)
(whiskers: 95% bootstrap over proteins)
S783
S2648
S8754
S669
S461
Ssym
Q3421
K3822
K2369
PTMUL
PTMUL-D
C RMSE hardly depends on the calibration set
calibrated on
scored on
S783 1.4 1.2 1.6 1.5 0.9 1.1 1.7 1.1 1.1 1.9 1.9
S2648 1.4 1.1 1.6 1.4 0.8 1.2 1.7 1.1 1.1 1.9 1.9
S8754 1.4 1.1 1.6 1.4 0.9 1.1 1.7 1.1 1.1 1.9 1.9
S669 1.5 1.1 1.6 1.4 0.9 1.2 1.7 1.1 1.1 1.9 1.9
S461 1.4 1.1 1.6 1.4 0.8 1.2 1.7 1.1 1.1 1.9 1.9
Ssym* 1.4 1.1 1.6 1.4 0.9 1.1 1.7 1.1 1.1 1.9 1.9
Q3421 1.4 1.1 1.6 1.4 0.8 1.2 1.7 1.1 1.1 1.9 1.9
K3822 1.4 1.1 1.6 1.4 0.9 1.1 1.7 1.1 1.1 1.9 1.9
K2369 1.4 1.2 1.6 1.4 0.9 1.1 1.7 1.1 1.1 1.9 1.9
PTMUL 1.6 1.2 1.7 1.5 1.0 1.2 1.7 1.2 1.2 1.9 2.0
PTMUL-D 1.5 1.1 1.6 1.4 0.9 1.2 1.7 1.1 1.1 1.9 1.9
D * calibrated on Ssym (and sets containing it, starred): not out-of-sample. Whiskers: 95% protein-level bootstrap
ESM-MSR (baseline; WT head, combined for multi-mutant sets)
MutateEverything st1
MutateEverything st2
ThermoMPNN
ThermoMPNN + ThermoMPNN-D
SPURS
SPURS multi
ESM-MSR before recalibration
RMSE (kcal/mol), recalibrated
S783
S2648
S8754
S669
S461
Ssym*
Q3421
K3822
K2369
PTMUL
PTMUL-D